Equivalence Relations

Definition

An ***equivalence relation*** on a set S, is a relation on S which is *reflexive, symmetric* and *transitive*.

Examples:

Let S = $\mathbb{Z}$ and define R = {(x,y) | x and y have the same parity} i.e., x and y are either both even or both odd.

The parity relation is an equivalence relation.

1. For any x $\in$ $\mathbb{Z}$, x has the same parity as itself, so (x,x) $\in$ R.
2. If (x,y) $\in$ R, x and y have the same parity, so (y,x) $\in$ R.
3. If (x,y) $\in$ R, and (y,z) $\in$ R, then x and z have the same parity as y, so they have the same parity as each other (if y is odd, both x and z are odd; if y is even, both x and z are even), thus (x,z) $\in$ R.

Examples

Let S = $\mathbb{R}$ and define the "square" relation $R = \{(x,y) \mid x^2 = y^2\}$.

The square relation is an equivalence relation.

1. For all $x \in \mathbb{R}$, $x^2 = x^2$, so $(x,x) \in R$.
2. If $(x,y) \in R$, $x^2 = y^2$, so $y^2 = x^2$ and $(y,x) \in R$.
3. If $(x,y) \in R$ and $(y,z) \in R$ then $x^2 = y^2 = z^2$, so $(x,z) \in R$.

For any set S, the identity relation on S, $I_S = \{(x,x) \mid x \in S\}$.

This is an equivalence relation for rather trivial reasons.

1. obvious
2. If $(x,y) \in R$ then $y = x$, so $(y,x) = (x,x) \in R$.
3. If $(x,y) \in R$ and $(y,z) \in R$ then $x = y = z$, so $(x,z) = (x,x) \in R$.

Modular Arithmetic

Let S = $\mathbb{Z}$. For each positive integer m, we define the ***modular relation*** $\equiv_m$,by $x \equiv_m y$ iff $m \mid (x-y)$, i.e. $\equiv_m = \{(x,y) : m \mid x - y\}$.

Examples:

$7 \equiv_5 2$, $11 \equiv_5 1$, $10 \equiv_5 0$, $-12 \equiv_5 3$

$7 \equiv_3 1$, $11 \equiv_3 2$, $10 \equiv_3 1$, $-12 \equiv_3 0$

Another way to think about the modular relation is:

$x \equiv_m y$ iff x and y have the same remainder when divided by m.

By the division algorithm, $x = mq_1 + r_1$, $y = mq_2 + r_2$, so

$x - y = m(q_1-q_2) + (r_1-r_2)$ so, $m \mid x-y$ iff $m \mid r_1 - r_2$. Since $|r_1-r_2| < m$,

$m|r_1 - r_2$ iff $r_1 - r_2 = 0$ iff $r_1 = r_2$.

Modular Arithmetic

Theorem: For any natural number m, the modular relation $\equiv_m$ is an equivalence relation on $\mathbb{Z}$.

Pf: For any x in $\mathbb{Z}$, since $x - x = 0$ and $m \mid 0$, $x \equiv_m x$. (Reflexitivity)

If $x \equiv_m y$ then $m \mid x - y$. Since $y - x = -(x-y)$, $m \mid y - x$, and so,

$y \equiv_m x$. (Symmetry)

If $x \equiv_m y$ and $y \equiv_m z$ then $m \mid x - y$ and $m \mid y - z$. Since

$$x - z = (x - y) + (y - z)$$

we have $m \mid x - z$, so $x \equiv_m z$. (Transitivity)

Equivalence Classes

Given an equivalence relation R on a set S, we define the ***equivalence class containing an element x*** of S by:

$$[x]_R = \{y \mid (x,y) \in R\} = \{y \mid x \, R \, y\}.$$

The text uses the notation x/R (which I am not fond of) for what I have called $[x]_R$.

Examples:

Let S = {1, 2, 3} and R = {(1,1), (2,2), (3,3), (1,2), (2,1)}.
Then [1] = {1,2} [2] = {1,2} [3] = {3}.

Let S = $\mathbb{R}$ and R = $\{(x,y) \mid x^2 = y^2\}$.
Then [0] = {0}, [1] = {1,-1}, [¼] = {¼, -¼}, [x] = {x, -x}.

Equivalence Classes

More Examples:

Let $S = \mathbb{Z}$ and $R = \{(x,y) \mid x \text{ and } y \text{ have the same parity}\}$.

$[0] = [2] = \ldots = [2k] = \{0, \pm2, \pm4, \pm6, \ldots, \pm2k, \ldots\}$

$[-1] = [1] = \ldots = [2k+1] = \{\pm1, \pm3, \pm5, \ldots, \pm2k+1, \ldots\}$

For any set S, $I_S = \{(x,x) \mid x \in S\}$.

$[a] = \{a\}$ for all $a \in S$.

Let $S = \mathbb{Z}$ and $R = " \equiv_5 "$.

$[0] = \{0, \pm5, \pm10, \pm15, \ldots, 5k\ \}$ $(k \in \mathbb{Z})$

$[1] = \{\ldots, -9, -4, 1, 6, 11, \ldots., 5k + 1\ \}$

$[2] = \{\ldots, -8, -3, 2,7,12, \ldots, 5k+2\ \}$

$[3] = \{\ldots, -7, -2, 3, 8, 13, \ldots, 5k+3\ \}$

$[4] = \{\ldots, -6, -1, 4, 9, 14, \ldots, 5k+4\ \}$

Properties of Equivalence Classes

Let R be an equivalence relation on the set S.

I. ***For all $x \in S$, $x \in [x]$.***

Since R is reflexive, $(x,x) \in R$ for all $x \in S$.

II. ***If $y \in [x]$ then $x \in [y]$, and $[x] = [y]$.***

Since R is symmetric, if $y \in [x]$ then $(x,y) \in R$ so $(y,x) \in R$ and we have $x \in [y]$. If $s \in [x]$, then $(x,s) \in R$, so $(s,x) \in R$ and then $(s,y) \in R$ (by transitivity) and finally $(y,s) \in R$, so $s \in [y]$. Similarly, if $t \in [y]$ then $t \in [x]$ and so, $[x] = [y]$.

III. ***For any x and $y \in S$, either $[x] = [y]$ or $[x] \cap [y] = \varnothing$.***

If there is a $z \in [x]$ which is also in $[y]$, then $(x,z) \in R$ and $(y,z) \in R$. By symmetry, $(z,y) \in R$ as well. By transitivity, $(x,y) \in R$, so $y \in [x]$. By II, $[x] = [y]$.

An Important Equivalence Relation

Let S be the set of fractions:

$$S = \left\{ \frac{p}{q} : \; p, q \in \mathbb{Z}, q \neq 0 \right\}$$

Define a relation R on S by:

$$\frac{a}{b} \; R \; \frac{c}{d} \quad \text{iff} \quad ad = bc.$$

This relation is an equivalence relation.

1) For any fraction a/b, a/b R a/b since ab = ba. (Reflexitivity)

2) If a/b R c/d, then ad = bc, so cb = da and c/d R a/b. (Symmetry)

3) If a/b R c/d, and c/d R e/f, then ad = bc and cf = de. Multiply the first equation by f, to get adf = bcf , so adf = bde. Divide by d (which is not 0) to get af = be, so a/b R e/f. (Transitivity)

An Important Equivalence Relation

The equivalence classes of this equivalence relation, for example:

$$\left[\frac{1}{1}\right] = \left\{\frac{2}{2}, \frac{3}{3}, \cdots, \frac{k}{k}, \cdots\right\}$$

$$\left[\frac{1}{2}\right] = \left\{\frac{2}{4}, \frac{3}{6}, \frac{4}{8}, \cdots, \frac{k}{2k}, \cdots\right\}$$

$$\left[\frac{4}{5}\right] = \left\{\frac{4}{5}, \frac{8}{10}, \frac{12}{15}, \cdots, 4\frac{k}{5}k, \cdots,\right\}$$

are called ***rational numbers***. The set of all the equivalence classes is denoted by $\mathbb{Q}$.

Partitions

A ***partition*** of a set S is a family *F* of *non-empty* subsets of S such that

(i) if A and B are in *F* then either A = B or $A \cap B = \varnothing$, and

(ii) $\underset{A \in F}{union}\, A = S$.

